College of Engineering – Department of Chemical Engineering

CURRICULUM

Bachelor of Science in Metallurgical Engineering

Academic Year 2025-2026

Reference CMOs: CMO N0. 100 s. 2017, CMO No. 4 s. 2018, CMO No. 20, s. 2013, CMO No. 39 s. 2021, and CMO No. 40 s. 2021

Curriculum Description

Metallurgical Engineering is a profession that involves the teaching and practice of physical and/or chemical processing and treatment of metals from ores to achieve properties required for their application. It is also the study of the properties of metallic materials by altering the microscopic structure with alloying additions and special treatments, and designing and processing techniques to transform these metals into useful shapes, such as billets, ingots, bars and slabs. In the mining and manufacturing industries, metallurgical engineers apply their knowledge of metals to solve complex problems. They deal with all aspects of the production and application of metals and their alloys. This means that they must learn about each stage of the metal-making process, from mining and extraction through alloy formation and the use of metals in finished products.

Program Education Objectives

The metallurgical engineering alumni three to five years after graduation shall:
PEO 1: Practice as specialist in solving complex metallurgical engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.
PEO 2: Assume leadership position in industry, academe, government, or private sector with consideration to social and ethical responsibility.
PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of

engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex engineering activities.

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings, and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
MATH 101 - Differential Calculus	4	0	4
MATH 102 - Integral Calculus	4	0	4
AI 102 - Probability and Statistics for AI	3	0	3
MATH 103 - Differential Equations	3	0	3
Subtotal	**14**	**0**	**14**
B. Natural/Physical Sciences			
CHEM 101 - Chemistry for Engineers	3	3	4
PHYS 101 - Physics for Engineers	3	3	4
CHEM 103 - Soil Chemistry	3	0	3
GeoE 202 - Principles of Geology	3	0	3
Subtotal	**12**	**6**	**14**
C. Basic Engineering Sciences			
ENGG 101 - Introduction to Engineering	0	3	1
ENGG 102 - Engineering Drawing	0	3	1
ENGG 103 - Computer-Aided Design	0	6	2
ENGG 104 - Engineering Economics	3	0	3
ENGG 105 - Technopreneurship	3	0	3
ENGG 106 - Engineering Management	3	0	3
ENGG 111 - Mechanics of Deformable Bodies	3	0	3
ENGG 120 - Mechanics of Rigid Bodies	4	0	4
CpE 101 - Computer Programming	0	3	1
Subtotal	**16**	**15**	**21**
D. Allied Courses			
MATH 105 - Calculus 3	2	0	2
CHEM 201 - Analytical Chemistry	2	6	4

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MetE 203 - Ore Mineralogy and Microscopy	2	6	4
MetE 205 - Principles of Mining	3	0	3
ENGG 112 - Fluid Mechanics	3	0	3
ENGG 118 - Materials Science and Engineering	3	0	3
MetE 311 - Workshop Theory, Practice and Process Safety	0	3	1
EE 214 - Basic Electrical Engineering	2	3	3
Subtotal	**17**	**18**	**23**
E. Professional Engineering Courses			
MetE 201 - Introduction to Metallurgical Engineering	1	0	1
MetE 302 - Metallurgical Analysis	1	6	3
MetE 206 - Metallurgical Physical Chemistry	3	3	4
MetE 207 - Mineral Processing 1	3	0	3
MetE 301 - Mineral Processing 2	3	6	5
MetE 304 - Hydrometallurgy	3	3	4
MetE 305 - Electrometallurgy	2	3	3
MetE 306 - Pyrometallurgy	3	3	4
MetE 404 - Industrial Production of Metals	3	0	3
MetE 303 - Mechanical and Adaptive Metallurgy	3	3	4
MetE 208 - Physical Metallurgy	3	3	4
MetE 309 - Metallurgical Evaluation	2	3	3
MetE 401 - Metallurgical Plant Design 1	2	0	2
MetE 402 - Metallurgical Plant Design 2	0	3	1
ENGG 116 - Methods of Research	3	0	3
MetE 308 - Computer Applications in Metallurgical Engineering	2	3	3
MetE 310 - Metallurgical Laws, Ethics, Seminars and Plant Visits	0	3	1
ENGG 117 - OJT/Internship (320 hrs)	320		4
MetE 307 - MetE Research Project 1	3	0	1
MetE 403 - MetE Research Project 2	0	3	1
MetE 405 - Integrative Studies	0	6	2
MetE 202 - Metallurgical Engineering Calculations	2	3	3
Subtotal	**362**	**54**	**62**
F. Technical Elective Courses			
MetEE 401 - Technical Elective 1 (Semiconductor Materials & Processes)	3	0	3
MetEE 402 - Technical Elective 2 (Special Topics in Metallurgical Engineering)	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL TECHNICAL COURSES	**427**	**93**	**140**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
GEd 101 - Understanding the Self	3	0	3
GEd 102 - Mathematics in the Modern World	3	0	3
GEd 104 - The Contemporary World	3	0	3
GEd 105 - Readings in Philippine History	3	0	3
GEd 106 - Purposive Communication	3	0	3
GEd 107 - Ethics	3	0	3
GEd 108 - Art Appreciation	3	0	3

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GEd 109 - Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Elective and Mandated Courses			
AI 101 - Linear Algebra for AI	3	0	3
AI 103 - Machine Learning and Neural Networks	3	0	3
GedE 101 - The Entrepreneurial Mind	3	0	3
GEd 103 - Life and Works of Rizal	3	0	3
ENGG 113 - Environmental Science and Engineering	3	0	3
Subtotal	**15**	**0**	**15**
C. Physical Activities Toward Health and Fitness (PATHFit)			
PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
PATHFit 4 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
NSTP 111 - National Service Training Program 1	3	0	3
NSTP 121 - National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**53**	**0**	**53**
GRAND TOTAL	**480**	**93**	**193**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	**14**
B. Natural/Physical Sciences	**14**
C. Basic Engineering Sciences	**21**
D. Allied Courses	**23**
E. Professional Engineering Courses	**62**
F. Technical Elective Courses	**6**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Elective and Mandate Courses	**15**
C. Physical Activities Toward Health and Fitness (PATHFit)	**8**
D. National Service Training Program	**6**
GRAND TOTAL	**193**

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CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN METALLURGICAL ENGINEERING
CMO No. 100 s. 2017, CMO No. 4 s. 2018, CMO No. 20 s. 2013, CMO No. 39 s. 2021 and CMO No. 40 s. 2021

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 101	Introduction to Engineering	1	0	3			TC - C
GEd 105	Readings in Philippine History	3	3	0			NTC - A
GEd 104	The Contemporary World	3	3	0			NTC - A
GEd 108	Art Appreciation	3	3	0			NTC - A
MATH 101	Differential Calculus	4	4	0			TC - A
GEd 102	Mathematics in the Modern World	3	3	0			NTC - A
NSTP 111	National Service Training Program 1	3	3	0			NTC - D
PATHFit 1	Movement Competency Training	2	2	0			NTC - C
CHEM 101	Chemistry for Engineers	4	3	3			TC - B
ENGG 102	Engineering Drawing	1	0	3			TC - C
	TOTAL	**27**	**24**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 101	Computer Programming	1	0	3			TC - C
GEd 101	Understanding the Self	3	3	0			NTC - A
GEd 106	Purposive Communication	3	3	0			NTC - A
GEd 109	Science, Technology and Society	3	3	0			NTC - A
MATH 102	Integral Calculus	4	4	0	MATH 101		TC - A
PHYS 101	Physics for Engineers	4	3	3	MATH 101	MATH 102	TC - B
GedE 101	The Entrepreneurial Mind	3	3	0			NTC - B
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC - D
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		NTC - C
	TOTAL	**26**	**24**	**6**			

FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GEd 103	Life and Works of Rizal	3	3	0			NTC - B
CHEM 103	Soil Chemistry	3	3	0			TC-B
GEd 107	Ethics	3	3	0			NTC - A
	TOTAL	**9**	**9**	**0**			

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SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
MetE 201	Introduction to Metallurgical Engineering	1	1	0			TC-E
MATH 103	Differential Equations	3	3	0	MATH 102		TC-A
MetE 202	Metallurgical Engineering Calculations	3	2	3	CHEM 101		TC-E
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101		TC-D
MetE 203	Ore Mineralogy and Microscopy	4	2	6	CHEM 101		TC-D
CHEM 201	Analytical Chemistry	4	2	6	CHEM 101		TC-D
ENGG 120	Mechanics of Rigid Bodies	4	4	0	MATH 102; PHYS 101		TC-C
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 1	2	2	0	PATHFit 1 & 2		NTC - C
AI 101	Linear Algebra for AI	3	3	0			NTC-B
	TOTAL	**27**	**17**	**15**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
MATH 105	Calculus 3	2	2	0	MATH 102		TC-D
MetE 205	Principles of Mining	3	3	0			TC-D
MetE 206	Metallurgical Physical Chemistry	4	3	3	CHEM 201		TC-E
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		TC-A
ENGG 111	Mechanics of Deformable Bodies	3	3	0	ENGG 120		TC-C
MetE 207	Mineral Processing 1	3	3	0	MetE 201, MetE 203		TC-E
MetE 208	Physical Metallurgy	4	3	3	ENGG 118		TC-E
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	2	0	PATHFit 1 & 2		NTC - C
	TOTAL	**24**	**22**	**6**			

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THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
MetE 301	Mineral Processing 2	5	3	6	MetE 207		TC-E
MetE 302	Metallurgical Analysis	3	1	6	CHEM 201		TC-E
MetE 303	Mechanical and Adaptive Metallurgy	4	3	3	MetE 208		TC-E
ENGG 103	Computer-Aided Design	2	0	6	ENGG 102		TC-C
MetE 304	Hydrometallurgy	4	3	3	MetE 206; CHEM 201		TC-E
ENGG 116	Methods of Research	3	3	0	3rd Year Standing, AI 102		TC-E
AI 103	Machine Learning and Neural Networks	3	3	0	AI 101, AI 102, CpE 101		NTC-B
	TOTAL	**24**	**16**	**24**			

THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 112	Fluid Mechanics	3	3	0	Fourth Year Standing		TC-D
MetE 305	Electrometallurgy	3	2	3	MetE 206; CHEM 201		TC-E
MetE 306	Pyrometallurgy	4	3	3	MetE 206		TC-E
MetE 307	MetE Research Project 1	1	3	0	ENGG 116; AI 102		TC-E
MetE 308	Computer Applications in Metallurgical Engineering	3	2	3	Third Year Standing		TC-E
MetE 309	Metallurgical Evaluation	3	2	3	MetE 303		TC-E
MetE 310	Metallurgical Law, Ethics, Seminars and Plant Visit	1	0	3	3rd Year Standing		TC-E
EE 214	Basic Electrical Engineering	3	2	3	PHYS 101		TC-D
ENGG 104	Engineering Economics	3	3	0	Third Year Standing, AI 102		TC-C
	TOTAL	**24**	**21**	**15**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GeoE 202	Principles of Geology	3	3	0	CHEM 103		TC-B
MetE 311	Workshop Theory, Practice and Process Safety	1	0	3			TC-D
ENGG 113	Environmental Science and Engineering	3	3	0	CHEM 101		NTC-B
	TOTAL	**7**	**6**	**3**			

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FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 117	OJT/Internship (320 hrs)	4	320 hrs		Fourth Year Standing		TC-E
MetE 401	MetE Plant Design 1	2	2	0	Fourth Year Standing	TC-E	
MetEE 401	MetE Elective 1	3	3	0		TC-F	
	TOTAL	**9**	**325**	**0**			

FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 105	Technopreneurship	3	3	0	Fourth Year Standing		TC-C
MetE 402	MetE Plant Design 2	1	0	3	Fourth Year Standing, MetE 401		TC-E
MetEE 402	MetE Elective 2	3	3	0	MetEE 401		TC-F
MetE 403	MetE Research Project 2	1	0	3	MetE 307		TC-E
ENGG 106	Engineering Management	3	3	0	ENGG 404		TC-C
MetE 404	Industrial Production of Metals	3	3	0	MetE 304, MetE 306, MetE 305		TC-E
MetE 405	Integrative Studies	2	0	6	Graduating Standing		TC-E
	TOTAL	**16**	**12**	**12**			